Recap + Assignment Check + If/Else Control Flow

Welcome back! Today we'll review what we've learned about variables and data types, check your assignment progress, and dive into one of the most important concepts in programming: control flow with if/else statements.

Recap: Variables & Data Types

Variables

Containers that store data you can reference and use throughout your program

Data Types

Define what kind of data a variable holds and how it's stored in memory

Common Data Types

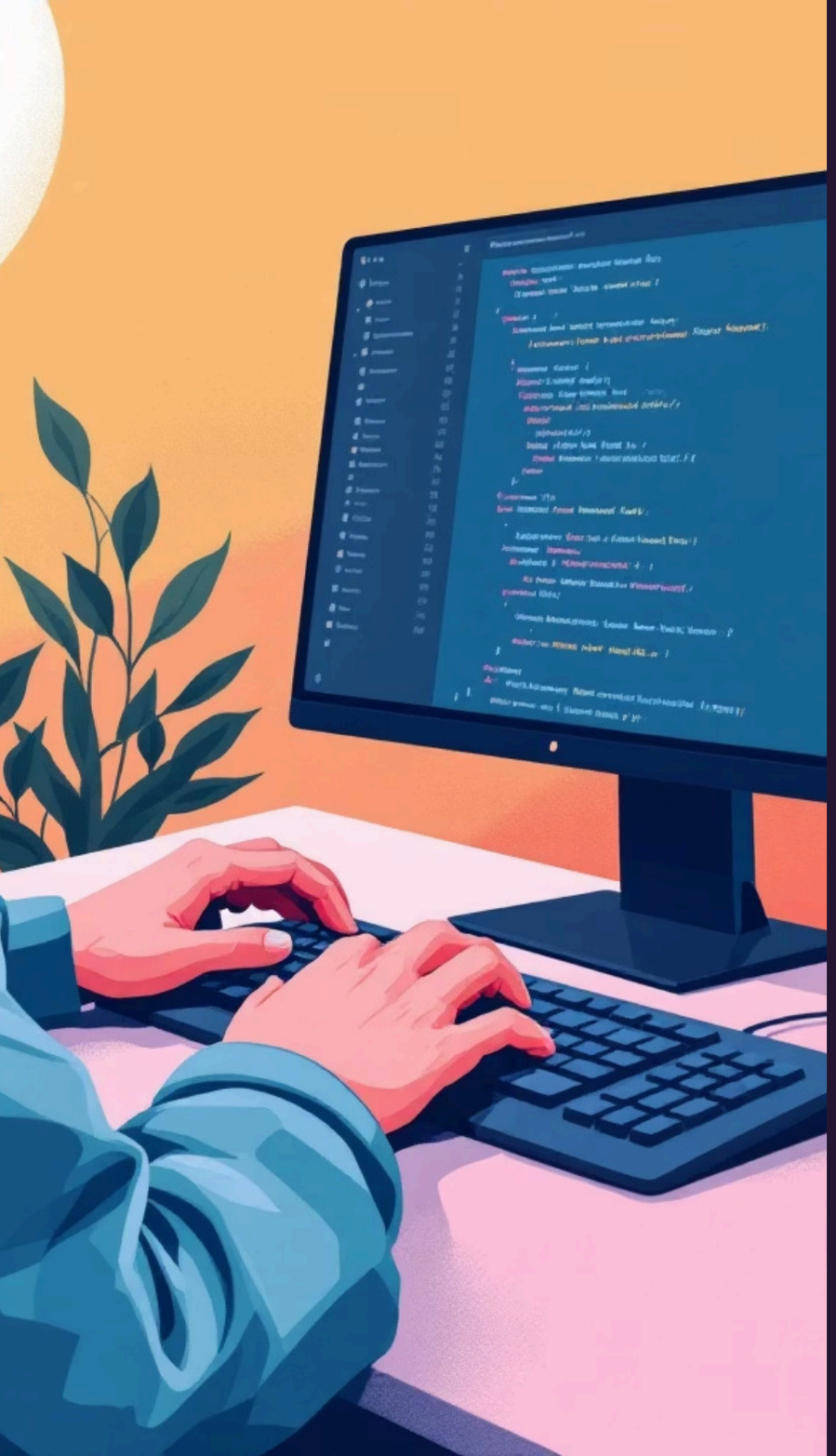
- **int** - Whole numbers (42, -7, 0)
- **float** - Decimals (3.14, -0.5, 2.0)
- **string** - Text in quotes ("Hello", 'Python')
- **bool** - True or False values

Example Code

```
name = "Arun"
age = 20
height = 5.7
is_student = True
```

Your Turn

Create 4 variables about yourself: your name, age, favorite number, and whether you're a student.



String Manipulation

Combining Strings

Strings are text data enclosed in quotes. You can combine them using the **+** operator or use f-string formatting for cleaner code.

```
name = "Arun"
print("My name is " + name)
print(f"My name is {name}")
```

Output:

My name is Arun
My name is Arun

Your Practice

Print your name and your city on the same line using both methods:

- Using **+** operator
- Using f-string formatting

Which method do you prefer and why?

Print Function Basics

Display Output

The **print()** function shows information to the user on the screen

Multiple Values

You can print several values at once by separating them with commas

Example Code

```
age = 21
print("Age:", age)
print("Hello", "World", "!")
```

Output

```
Age: 21
Hello World !
```

Your Task: Print your school name and current grade level on the same line.

Input Function & Type Casting

Getting User Input

The **input()** function lets users enter data while your program runs. It always returns a string, so you need to convert it using **int()** or **float()** for calculations.

01

Call input()

Display prompt and wait for user

02

Store as string

Input is captured as text

03

Convert type

Use int() or float() to change data type

Example

```
age = int(input("Enter age: "))
print("Your age is", age)
```

Output:

```
Enter age: 20
Your age is 20
```

Your Turn: Take two numbers from user input and print their sum.

Assignment Check: Mini Practice

Original Code

```
a = int(input("Enter number: "))
b = int(input("Enter number: "))
print("Sum =", a + b)
```

Modify This Code

Now that you've reviewed variables, strings, and input, let's practice basic arithmetic operations. For each modification below, run the code and test with different numbers:

Subtraction

Calculate the difference between a and b

Multiplication

Find the product of a and b

Division

Divide a by b (watch for division by zero!)

Control Statements: The Decision Makers

Control flow statements let your program make decisions based on conditions. Think of them as the brain of your program, deciding which path to take next.

`if`

Runs code if condition is True

`elif`

Checks additional conditions

`else`

Runs if no conditions match

These three work together to create branching logic that makes programs smart and responsive to different situations.

If Statement: The Basic Condition

How It Works

An **if** statement checks whether a condition is True. If it is, the indented code block below runs. If the condition is False, the program skips that block entirely.

Example Code

```
age = 18
if age >= 18:
    print("Eligible to vote")
    print("Register to vote!")
```

Output

```
Eligible to vote
Register to vote!
```

Key Points

- Condition must evaluate to True/False
- Colon **:** starts the block
- Indentation shows which code belongs
- Multiple lines can be in the block

Your Task: Write a program that checks if a number is positive (greater than 0). Test with both positive and negative numbers.

If-Else: Two Paths

The Complete Pattern

An **if-else** statement gives you two options: if the condition is True, run the first block. Otherwise, run the second block. This is perfect for binary decisions.

If Block

Runs when condition is True

Else Block

Runs when condition is False

Example: Even or Odd

```
num = int(input("Enter number: "))
if num % 2 == 0:
    print("Even")
else:
    print("Odd")
```

Output:

```
Enter number: 4
Even
```

Your Task: Write a program that checks if a student passed (marks >= 40). Print "Passed" or "Failed" accordingly.

If-Elif-Else & Operators

Multiple Conditions

Use **elif** (else if) when you have more than two possible outcomes. The program checks each condition in order until one is True, then skips the rest.

Example: Grading System

```
marks = int(input("Enter marks: "))
if marks >= 90:
    print("Grade A")
elif marks >= 75:
    print("Grade B")
elif marks >= 60:
    print("Grade C")
else:
    print("Grade D")
```

Comparison Operators

== (equal)	> (greater)	>= (greater/equal)
!= (not equal)	< (less)	<= (less/equal)

Logical Operators

- **and** - Both conditions must be True
- **or** - At least one condition must be True
- **not** - Inverts the condition

Example with Logical Operators

```
age = 20
citizen = True
if age >= 18 and citizen:
    print("Eligible to vote")
```

Your Challenge: Write a program that checks if a user can enter a club (age >= 21 AND has ID proof). Use input() for both values.